LAFAYETTE COLLEGE

Simulation of Semi-deformable Crop Structures for Robotic Harvesting

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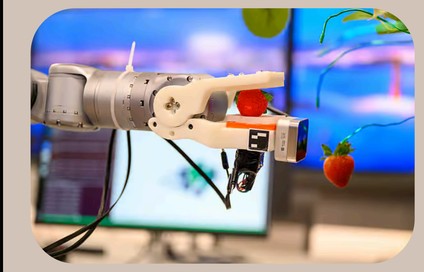
Motivation



- Strawberry harvesting robots often rely on high technological complexity to address delicate fruit handling, including touch sensors, pedicel localization, and 6-DOF robotic arms.

Such complexity constrains economic feasibility by increasing system cost, integration difficulty, and operational requirements.

Human picking suggests simpler interactions may work better

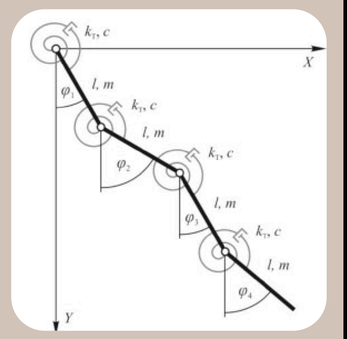
Cage fingers around the berry while allowing the pedicel to slide between the fingers.

A simulation testbed is needed to evaluate strawberry–robot interactions and support the design of economically feasible harvesting systems.

Results

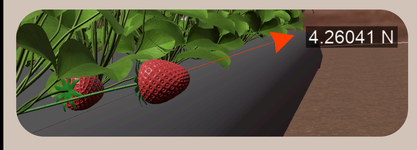
Photorealistic Crop Environment for Future Computer Vision Integration

Modeling Pedicel Deformation in Webots Using Rigid Bodies, Ball Joints, Springs, and Dampers



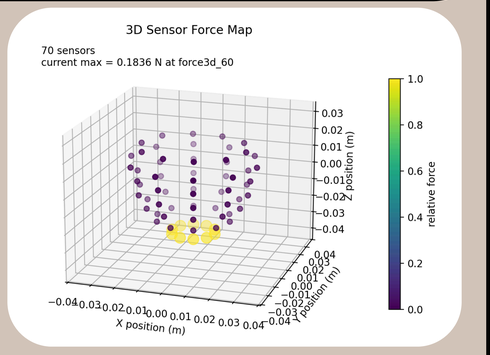
- PROTO-configurable link count, length, diameter, joint damping, and spring stiffness

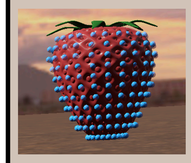
Fruit Abscission Using Webots Connector Nodes



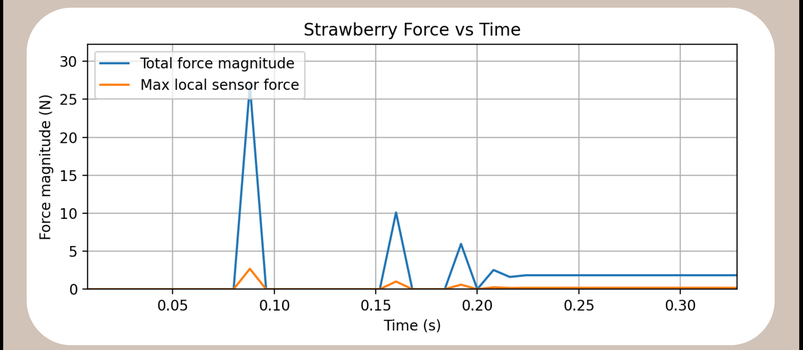
- Configurable tensile and shearing threshold for abcission

Distributed Force Sensing with Live Visualization





- PROTO-configurable touch sensor quantity, positioning, size, etc.



Future Work

- Validate stem stiffness, damping, and fruit detachment force with physical tests
- Tune Webots parameters to match real strawberry motion
- Compare cage-style end effectors against more complex harvesting approaches
- Use force maps to evaluate damage risk during harvesting